BANG Meeting

Ian Snider

BANG Updates
10.31.2025

Bateman

$$\dot{N}_p = R - \lambda N_p$$

$$\dot{N}_d = \lambda_p N_p - \lambda_d N_d$$

$N_p(0) = N_d(0) = 0$

`scipy.integrate.solve_ivp`

$^{147}\mathrm{Ba}\ t_{\frac{1}{2}} = 0.894\ \mathrm{s}$ $^{147}\mathrm{La}\ t_{\frac{1}{2}} = 4.06\ \mathrm{s}$
$^{148}\mathrm{Ba}\ t_{\frac{1}{2}} = 0.619\ \mathrm{s}$ $^{148}\mathrm{La}\ t_{\frac{1}{2}} = 1.411\ \mathrm{s}$

maximize $\mathrm{SNR}(t_{\mathrm{cycle}}) = \frac{\int_0^{t_{\mathrm{cycle}}} A_p \,\mathrm{d}t}{\int_0^{t_{\mathrm{cycle}}} A_d \,\mathrm{d}t}$ might have to specify minimum contamination